Many-body expansion of the CT commutators

generated by `expand_commutators.py`

All commutator blocks of the `commutator` stage, taken from its API (`commutator_blocks()` of `commutator/generate_commutators.py`) and expanded with the generalized Wick theorem (`generalized_wick_th`
The truncation is by perturbation order (Møller–Plesset counting $\hat{h} \sim O(1)$, $\hat{V} \sim O(\lambda)$, $\hat{T} \sim O(\lambda)$; the Fock operator differs from $\hat{h}$ only at $O(\lambda)$), correct through $O(\lambda^2)$: the full first-order term $[\hat{H}_b, \hat{A}]$ with $\hat{H}_b = \hat{h} + \hat{V}$, and the Fock half of the second order, $\frac{1}{2}[[\hat{F}, \hat{A}], \hat{A}]$ (writing $\hat{F}$ for $\hat{h}$ there changes the result only at $O(\lambda^3)$). The valence double commutator is $O(\lambda^3)$ and is not expanded. Repeated indices are summed over their space. The index spaces and the delta rules are those of `readme.md`:

$$p_1, p_2, \ldots : \text{ whole space}, \qquad i_1, i_2, \ldots : \text{ occupied}, \qquad a_1, a_2, \ldots : \text{ virtual}, \tag{1}$$

$$\delta(\text{whole}, \text{vir}) \to \text{virtual}, \qquad \delta(\text{whole}, \text{occ}) \to \text{occupied}, \qquad \delta(\text{occ}, \text{vir}) = 0. \tag{2}$$

The operators are

$$\hat{h} = h^q_p \, \hat{E}^p_q, \qquad \hat{V} = \tfrac{1}{2} g^{pq}_{rs} \, \hat{E}^{rs}_{pq}, \tag{3}$$

$$\hat{F} = f^q_p \, \hat{E}^p_q, \qquad \hat{T}_1 = t^a_i \, \hat{E}^a_i, \tag{4}$$

$$\hat{T}_2 = \tfrac{1}{2} t^{ab}_{ij} \, \hat{E}^{ab}_{ij}, \tag{5}$$

with $\hat{h}$, $\hat{F}$ and $\hat{V}$ Hermitian. The dummy indices are numbered per space ($a_1, a_2, \ldots$ virtual, $i_1, i_2, \ldots$ occupied, $p_1, p_2, \ldots$ whole space); the second excitation of the nested commutators carries the next pair (a_3, a_4 and i_3, i_4). The coefficients are real and follow the spin-integrated convention: no index antisymmetry is used anywhere, g^{pq}_{rs} and t^{ab}_{ij} carry only the simultaneous pair swap $g^{pq}_{rs} = g^{qp}_{sr}$, $t^{ab}_{ij} = t^{ba}_{ji}$, and $h^q_p = h^p_q$, $f^q_p = f^p_q$, $g^{pq}_{rs} = g^{rs}_{pq}$. Conjugate blocks are expanded by conjugating the first-order result term by term ($\hat{E}^{u\cdots}_{l\cdots} \to \hat{E}^{l\cdots}_{u\cdots}$, $f^q_p \to f^p_q$, $g^{pq}_{rs} \to g^{rs}_{pq}$; t is real and unchanged). Blocks containing operators of rank higher than two are marked; their cumulant expansion (and the $[\ldots]_{1,2}$ truncation) is done separately. The expanded blocks are available programmatically through `expanded_blocks()`.

Equal terms are merged once per block, after the full expansion (`merge`), over every identity that is exact in this convention (`readme.md`, simplification steps 2–6): the diagonal pair-permutation symmetry of the spin-summed $\hat{E}$ (the two index rows may only be permuted together, with no sign, $\hat{E}^{qp\cdots}_{rs\cdots} = \hat{E}^{pq\cdots}_{sr\cdots}$; a single row alone is *not* a symmetry of the spin-summed operator), the pair symmetries of g and t above, and the renaming of summed indices — the free indices of the operator are the summation labels of the operator basis sum, so they can be renamed like tensor dummies.

Every block is verified numerically against the direct nested commutator on Jordan–Wigner matrices, twice: with spin-orbital operators (4 modes, 2 occupied + 2 virtual) and with the genuinely spin-summed operators (4 spatial orbitals = 8 spin orbitals); the second check is the one that sees the merging steps.

1 First-order commutators

$$\left[\hat{h},\,\hat{T}_1\right] = -h^{i_1}_{p_1} t^{a_1}_{i_1} \hat{E}^{a_1}_{p_1} + h^{a_1}_{p_1} t^{a_1}_{i_1} \hat{E}^{p_1}_{i_1} \tag{6}$$

$$\left[\hat{h},\,\hat{T}_2\right] = -h^{i_2}_{p_1} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 p_1} \tag{7}$$

$$+ h^{a_2}_{p_1} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{8}$$

$$\left[\hat{V},\,\hat{T}_1\right] = -g^{i_1 p_1}_{p_2 p_3} t^{a_1}_{i_1} \hat{E}^{a_1 p_1}_{p_2 p_3} \tag{9}$$

$$+ g^{a_1 p_3}_{p_1 p_2} t^{a_1}_{i_1} \hat{E}^{p_1 p_2}_{i_1 p_3} \tag{10}$$

$$\left[\hat{V},\,\hat{T}_2\right] = -\tfrac{1}{2}\, g^{i_1 i_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{p_1 p_2} \tag{11}$$

$$+ \tfrac{1}{2}\, g^{a_1 a_2}_{p_1 p_2} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{p_1 p_2}_{i_1 i_2} \tag{12}$$

$$- g^{i_2 p_1}_{p_2 p_3} t^{a_1 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2 p_1}_{i_1 p_2 p_3} \tag{13}$$

$$+ g^{a_2 p_3}_{p_1 p_2} t^{a_2 a_1}_{i_2 i_1} \hat{E}^{a_1 p_1 p_2}_{i_1 i_2 p_3} \quad \text{(three-body terms)} \tag{14}$$

2 Second order: Fock blocks

$$\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_1\right] = -2\, f^{a_2}_{i_2} t^{a_1}_{i_2} t^{a_2}_{i_1} \hat{E}^{a_1}_{i_1} \tag{15}$$

$$\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_2\right] = -f^{a_3}_{i_3} t^{a_1 a_2}_{i_1 i_3} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{16}$$

$$- f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{17}$$

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_1\right] = -f^{a_3}_{i_3} t^{a_1 a_2}_{i_1 i_3} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{18}$$

$$- f^{a_3}_{i_3} t^{a_1}_{i_3} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 a_2}_{i_1 i_2} \tag{19}$$

$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right] = -2\, f^{a_4}_{i_4} t^{a_1 a_2}_{i_1 i_4} t^{a_4 a_3}_{i_2 i_3} \hat{E}^{a_1 a_2 a_3}_{i_1 i_2 i_3} \quad \text{(three-body terms)} \tag{20}$$

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_1\right] = f^{i_1}_{i_2} t^{a_1}_{i_1} t^{a_2}_{i_2} \hat{E}^{a_1}_{a_2} \tag{21}$$

$$- f^{a_2}_{p_1} t^{a_1}_{i_1} t^{a_2}_{i_1} \hat{E}^{a_1}_{p_1} + f^{a_1}_{a_2} t^{a_1}_{i_1} t^{a_2}_{i_2} \hat{E}^{i_1}_{i_2} \tag{22}$$

$$- f^{i_2}_{p_1} t^{a_1}_{i_1} t^{a_1}_{i_2} \hat{E}^{p_1}_{i_1} \tag{23}$$

$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right] = f^{i_2}_{i_3} t^{a_1 a_2}_{i_2 i_1} t^{a_3}_{i_3} \hat{E}^{a_1 a_2}_{a_3 i_1} \tag{24}$$

$$- f^{a_3}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_3}_{i_2} \hat{E}^{a_1 a_2}_{i_1 p_1} \tag{25}$$

$$+ f^{a_2}_{a_3} t^{a_2 a_1}_{i_3 i_2} t^{a_3}_{i_1} \hat{E}^{a_1 i_1}_{i_2 i_3} \tag{26}$$
$$- f^{i_3}_{p_1} t^{a_2 a_1}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{a_1 p_1}_{i_1 i_2} \tag{27}$$

$$\left[\left[\hat{F},\, \hat{T}_2\right]^\dagger,\, \hat{T}_1\right] = f^{i_2}_{i_3} t^{a_1}_{i_2} t^{a_2 a_3}_{i_3 i_1} \hat{E}^{a_1 i_1}_{a_2 a_3} \tag{28}$$
$$- f^{a_3}_{p_1} t^{a_1}_{i_2} t^{a_3 a_2}_{i_1 i_2} \hat{E}^{a_1 i_1}_{a_2 p_1} \tag{29}$$
$$- f^{a_3}_{p_1} t^{a_1}_{i_2} t^{a_3 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1}_{p_1 a_2} \tag{30}$$
$$+ f^{i_1}_{p_1} t^{a_1}_{i_2} t^{a_2 a_3}_{i_2 i_1} \hat{E}^{a_1 p_1}_{a_2 a_3} \tag{31}$$
$$+ f^{a_2}_{a_3} t^{a_2 a_1}_{i_2 i_1} t^{a_3}_{i_3} \hat{E}^{i_1 i_2}_{a_1 i_3} \tag{32}$$
$$+ f^{a_1}_{p_1} t^{a_1 a_2}_{i_2 i_1} t^{a_2}_{i_3} \hat{E}^{i_1 i_2}_{i_3 p_1} \tag{33}$$
$$- f^{i_3}_{p_1} t^{a_2 a_1}_{i_3 i_1} t^{a_2}_{i_2} \hat{E}^{i_1 p_1}_{a_1 i_2} \tag{34}$$
$$- f^{i_3}_{p_1} t^{a_2 a_1}_{i_1 i_3} t^{a_2}_{i_2} \hat{E}^{i_1 p_1}_{i_2 a_1} \tag{35}$$

$$\left[\left[\hat{F},\, \hat{T}_2\right]^\dagger,\, \hat{T}_2\right] = f^{i_1}_{i_2} t^{a_1 a_2}_{i_1 i_3} t^{a_3 a_4}_{i_2 i_3} \hat{E}^{a_1 a_2}_{a_3 a_4} \tag{36}$$
$$- f^{a_4}_{p_1} t^{a_1 a_2}_{i_1 i_2} t^{a_4 a_3}_{i_2 i_1} \hat{E}^{a_1 a_2}_{a_3 p_1} \tag{37}$$
$$+ f^{a_1}_{a_2} t^{a_1 a_3}_{i_1 i_2} t^{a_2 a_3}_{i_3 i_4} \hat{E}^{i_1 i_2}_{i_3 i_4} \tag{38}$$
$$- f^{i_4}_{p_1} t^{a_1 a_2}_{i_1 i_4} t^{a_1 a_2}_{i_2 i_3} \hat{E}^{i_1 p_1}_{i_2 i_3} \tag{39}$$
$$+ f^{i_3}_{i_4} t^{a_1 a_2}_{i_3 i_2} t^{a_3 a_4}_{i_4 i_1} \hat{E}^{a_1 a_2 i_1}_{a_3 i_2 a_4} \tag{40}$$
$$- f^{a_4}_{p_1} t^{a_1 a_2}_{i_3 i_2} t^{a_4 a_3}_{i_1 i_3} \hat{E}^{a_1 a_2 i_1}_{a_3 i_2 p_1} \tag{41}$$
$$- f^{a_4}_{p_1} t^{a_1 a_2}_{i_2 i_3} t^{a_4 a_3}_{i_3 i_1} \hat{E}^{a_1 a_2 i_1}_{i_2 p_1 a_3} \tag{42}$$
$$+ f^{i_2}_{p_1} t^{a_1 a_2}_{i_3 i_1} t^{a_3 a_4}_{i_3 i_2} \hat{E}^{a_1 a_2 p_1}_{a_3 i_1 a_4} \tag{43}$$
$$+ f^{a_3}_{a_4} t^{a_3 a_1}_{i_4 i_3} t^{a_4 a_2}_{i_2 i_1} \hat{E}^{a_1 i_1 i_2}_{i_3 a_2 i_4} \tag{44}$$
$$+ f^{a_2}_{p_1} t^{a_2 a_3}_{i_2 i_1} t^{a_3 a_1}_{i_4 i_3} \hat{E}^{a_1 i_1 i_2}_{i_3 i_4 p_1} \tag{45}$$
$$- f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_4 i_1} \hat{E}^{a_1 i_1 p_1}_{i_2 a_2 i_3} \tag{46}$$
$$- f^{i_4}_{p_1} t^{a_3 a_1}_{i_3 i_2} t^{a_3 a_2}_{i_1 i_4} \hat{E}^{a_1 i_1 p_1}_{i_2 i_3 a_2} \quad \text{(three-body terms)} \tag{47}$$

3 Assembly

With the blocks above, the transcorrelated Hamiltonian through $O(\lambda^2)$ is

$$\left[\hat{H}_b,\, \hat{A}\right] = \left[\hat{h},\, \hat{A}\right] + \left[\hat{V},\, \hat{A}\right], \tag{48}$$
$$\left[\hat{h},\, \hat{A}\right] = \left[\hat{h},\, \hat{T}_1\right] + \left[\hat{h},\, \hat{T}_2\right] + \text{h.c.}, \tag{49}$$
$$\left[\hat{V},\, \hat{A}\right] = \left[\hat{V},\, \hat{T}_1\right] + \left[\hat{V},\, \hat{T}_2\right] + \text{h.c.}, \tag{50}$$
$$\tfrac{1}{2}\left[\left[\hat{F},\, \hat{A}\right],\, \hat{A}\right] = \tfrac{1}{2}\sum_{r,s=1}^{2}\left(\left[\left[\hat{F},\, \hat{T}_r\right],\, \hat{T}_s\right] + \left[\left[\hat{F},\, \hat{T}_r\right]^\dagger,\, \hat{T}_s\right]\right) + \text{h.c.}, \tag{51}$$

the valence double commutator $\frac{1}{2}[[\hat{V}, \hat{A}], \hat{A}] \sim O(\lambda^3)$ being dropped by the perturbation-order truncation.